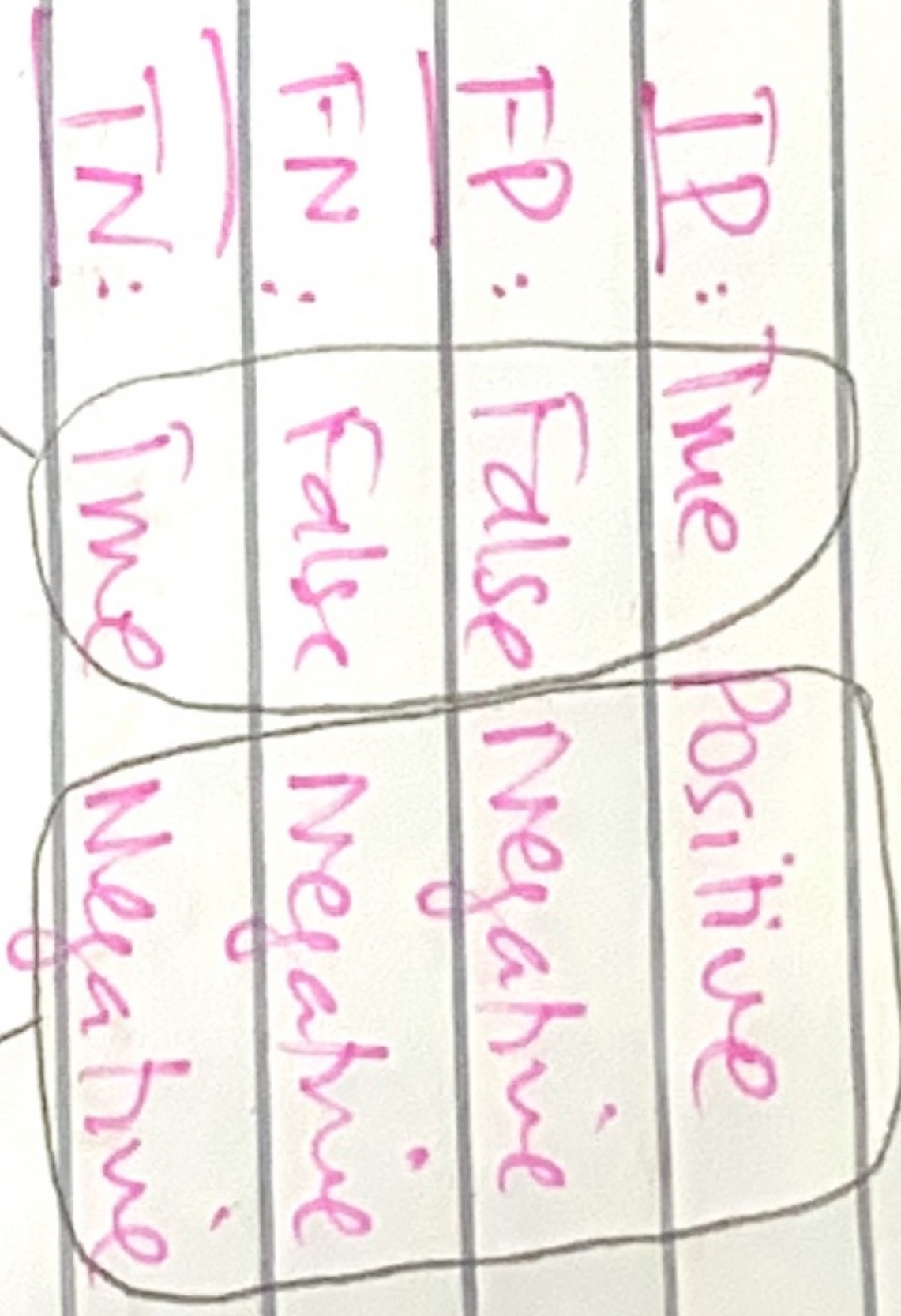


LECTURE-20

Performance Measures - Confusion Matrix:

- Confusion Matrix is measurement technique for performance in ML
- It's a $N \times N$ matrix where N is no. of target classes.

		Actual values	
Predicted Values	Result = 1 (Positive)	Result = 1 (Positive)	Result = 0 (Negative)
	Result = 0 (Negative)	TP	FP
		FN	TN



Tells if the results predicted are same as Actual (T) or Not (F)

Tells what did the model predicted

Positive = 1

Negative = 0

e.g: FP means model predicted = 1, actual = 0.

Q: ID	Actual Sick	Predicted Sick	Outcome
1	1	1	TP
2	0	0	TN
3	0	0	TN
4	1	1	TP
5	0	0	TN
6	0	0	TN
7	1	0	FN
8	0	1	FP
9	0	0	TN
10	1	0	FN

		Actual		
		1	0	
Predicted	1	2 (TP)	1 (FP)	3
	0	2 (FN)	5 (TN)	7
		4	6	10

⇒ Accuracy: No. of all correct prediction including result = 0 & 1 (TN & TP) divided by total dataset.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

(Above res).

$$\text{Accuracy} = \frac{2 + 5}{10} = 70\%$$

Model can predict ~~the~~ people being sick or not 70% of the time

Problem: If dataset is very large ^{like 1000 ppl} & outlier that only some people are sick (like 30/40) then accuracy = 96% if model predicted 30/40 from those 1000 people to be sick which were actually sick.

		Actual	
		1	0
Predicted	1	30 (TP)	30 (FP)
	0	10 (FN)	930 (TN)

As we're very much interested in finding sick people to cure them so when model predicts 10 people in FN ~~that~~ ~~are~~ ~~actual~~ to be healthy But they're actually sick; So that's a big problem in accuracy.

For that we'll use RECALL.

Sensitivity

(True, result = 1 $\rightarrow$ TP)

$\Rightarrow$ RECALL: No. of correct positive predictions $\uparrow$ divided by Actual total positive ~~prediction~~ (whether result = 0 or 1).

$$REC = \frac{TP}{TP + FN}$$

$\nearrow$ of pencil table on prev. page.

$$Rec = \frac{30}{30 + 10} = 75\%$$

75% of +ve were successfully predicted as +ve by model.

$\Rightarrow$ Precision: No. of correct positive predictions (TP) divided by total no. of positive predictions (1st row $\rightarrow$ TP + FP).

$$Precision = \frac{TP}{TP + FP}$$

$$Precision = \frac{30}{30 + 30} = 50\%$$

- 50% of correctly predicted cases turned out to be positive cases.
- This tells reliability.

$\Rightarrow$ F1 Score: Harmonic mean of "precision and recall" balancing these two. Useful for imbalanced dataset.

$$F1 \text{ Score} = 2 \left(\frac{Precision \times Recall}{Precision + Recall} \right)$$

$$F1 \text{ Score} = 2 \times \frac{0.75 \times 0.50}{0.75 + 0.5} = \boxed{0.60}$$

⇒ Specificity/~~True~~ TN Rate: Measures the portion of actual negative identified by model correct. how well model avoids FP.

$$\text{TN Rate} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

⇒ Matthews Correlation: Balance measure.

+1: Perfect Predictions, 0: random guesses, -1: Completely wrong.

$$\text{MCC} = \frac{(\text{TP} \cdot \text{TN}) - (\text{FP} \cdot \text{FN})}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}$$

⇒ ROC/AUC: The ability of classifier to distinguish b/w classes plotting graph of TP against FP. (Ideally classifier has AUC close to 1)

NOTE: For Regression Performance Measurement; we've four metrics :-
MAE, MSE, RMSE, R^2 (all done in lecture 15)
